

Mark schemes

Q1.

- (a) (Gradient = $10 - 6 + 5$) = 9

correct gradient from sub $x = 1$ into $\frac{dy}{dx}$

B1

$$y - 4 = \text{"their 9"} (x - 1)$$

or $y = \text{"their 9"} x + c$ **and** attempt

to find c using $x = 1$ and $y = 4$

}

must attempt to use given expression for $\frac{dy}{dx}$ **and** must be attempting tangent and not normal **and** coordinates must be correct

M1

$$y = 9x - 5$$

condone $y = 9x + c, \dots$ $c = -5$

A1

3

- (b) $(y =) \frac{10}{5}x^5 - \frac{6}{3}x^3 + 5x + C$
one term correct

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M1

another term correct

A1

all integration correct including $+ C$

A1

$$4 = 2 - 2 + 5 + C \Rightarrow C = -1$$

substituting both $x = 1$ and $y = 4$ **and** attempting to find C

m1

$$y = 2x^5 - 2x^3 + 5x - 1$$

must have $y = \dots$ and coefficients simplified

Q2.

(a) $h = 1$

PI

B1

$$f(x) = \frac{1}{x^2 + 1}$$

$$I \approx \frac{h}{2} \{f(1) + f(5) + 2[f(2) + f(3) + f(4)]\}$$

$$\frac{h}{2} \{f(1) + f(5) + 2[f(2) + f(3) + f(4)]\}$$

OE summing of areas of the four 'trapezia'...

M1

$$\frac{h}{2} \text{ with } \{...\} = \frac{1}{2} + \frac{1}{26} + 2\left(\frac{1}{5} + \frac{1}{10} + \frac{1}{17}\right)$$

$$= 0.5 + 0.03(84...) + 2[0.2 + 0.1 + 0.05(88...)]$$

$$= 0.538(46...) + 2[0.358(82...)] = 1.256(108...)$$

OE Accept 2dp (rounded or truncated) for non-terminating decs. equiv.

A1

$$(I \approx) 0.628054... = \frac{694}{1105} = 0.628 \text{ (to 3sf)}$$

CAO Must be 0.628

SC for those who use 5 strips, max possible is B0M1A1A0

A1

4

(b) (i) $\int \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx = \frac{x^{-\frac{1}{2}}}{-1/2} + \frac{6x^{\frac{3}{2}}}{3/2} (+ c)$

One term correct (even unsimplified)

M1

Both terms correct (even unsimplified)

A1

$$= -2x^{-0.5} + 4x^{1.5} (+ c)$$

Must be simplified

A1

3

(ii) $\int_1^4 \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx = [-2(4^{-0.5}) + 4(4^{1.5})] - [-2(1^{-0.5}) + 4(1^{1.5})]$

Attempt to calculate $F(4) - F(1)$ where $F(x)$ follows integration and is not just the integrand

M1

$$= (-1 + 32) - (-2 + 4) = 29$$

Since 'Hence' NMS scores 0 / 2

A1

2

[9]

Q3.

(a) (i) $\{(2 + y)^3 =\} 8 + 12y + 6y^2 + y^3$
At least 3 terms simplified and correct

M1

All correct

A1

2

(ii) $(2 + x^{-2})^3 = 8 + 12x^{-2} + 6(x^{-2})^2 + (x^{-2})^3$

A replacement of y by x^{-2} in c's (a)(i) working. Pl

M1

$$(2 - x^{-2})^3 = 8 - 12x^{-2} + 6(x^{-2})^2 - (x^{-2})^3$$

Ft one incorrect coefficient in (a)(i) expansion.

A1F

$$(2 + x^{-2})^3 + (2 - x^{-2})^3 = 16 + 12x^{-4}$$

CSO Be convinced.

SC2 for a fully correct solution, not using 'Hence'

A1

3

(b) (i) $\int \left[(2 + x^{-2})^3 + (2 - x^{-2})^3 \right] dx = 16x - 4x^{-3} (+c)$

Valid method to obtain the correct power of x after integrating qx^{-4} .

M1

$16x - 4x^{-3}$ or $16x - 4/x^3$ condone missing '+c'.

Ft on c's p and q values. Coefficients and signs must be simplified

A1F

2

(ii) $\int_1^2 \dots dx = [16(2) - 4(2^{-3})] - [16 - 4]$
F(2) - F(1) following integration (b)(i)

M1

$= 31.5 - 12 = 19.5$

*OE Ft on c's **positive integer** values of p and q.*

Since 'Hence' NMS scores 0/2

A1F

2

[9]

Q4.

(a) $\frac{dy}{dx} = 12 - 5x^{\frac{2}{3}}$
 $kx^{\frac{2}{3}}$ term.

ACF

M1


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A1

2

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(b) (i) When $x = 0$, $\frac{dy}{dx} = 12$

Ft on c's y' evaluated correctly at $x = 0$

B1F

Eqn of tangent at O is $y = 12x$

OE Ft on c's value for $y'(0)$ provided $y'(0) > 0$.

B1F

2

(ii) When $x = 8$, $\frac{dy}{dx} = 12 - 5 \times (8)^{\frac{2}{3}}$
Attempt to find $\frac{dy}{dx}$ when $x = 8$

M1

Equation of tangent at (8, 0) is $y - 0 = y'(8)[x - 8]$

$$y = y'(8)[x - 8] \text{ OE}$$

M1

$$y = -8(x - 8) \Rightarrow y + 8x = 64$$

CSO AG

A1

3

$$(c) \quad \int \left(12x - 3x^{\frac{5}{3}} \right) dx = \frac{12x^2}{2} - \frac{3x^{\frac{8}{3}}}{\frac{8}{3}} (+c)$$

$kx^{\frac{5}{3}+1}$ term after integrating, condone k left unsimplified for this M mark.

$$= 6x^2 - \frac{9}{8}x^{\frac{8}{3}} (+c)$$

For $6x^2$ OE eg $(12x^2 / 2)$

B1

For $-\frac{9}{8}x^{\frac{8}{3}}$ OE

A1

3

(d) Area bounded by curve and x -axis

$$= \int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx = 6 \times 8^2 - \frac{9}{8} \times (8)^{\frac{8}{3}}$$

$\pm F(8) \{- F(0)\}$ PI following integration

M1

$$= 384 - 288 = 96$$

PI by correct final answer if evaluation not seen here

A1

At P , $12x + 8x = 64$

Solving $y + 8x = 64$ and c 's $y = kx$, $k > 0$, down to an eqn in one variable...

$$[y + 2y / 3 = 64]$$

M1

$$(x_p = 3.2) \quad y_p = 38.4$$

For $y_p 38.4$ OE [If using integration to find area of triangle, award A1 if both ' $x_p = 3.2$ ' and correct integration of correct eqns of the 2 lines]

A1

$$\text{Area of triangle } OPA = \frac{1}{2} \times 8 \times y_p$$

OE Need perpendicular ht to be linked to $y_p > 0$.

M1

$$\text{Area of shaded region} = \text{Area } \Delta OPA - \int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx$$

M0 if evaluated to a value < 0

M1

$$= 153.6 - 96 = 57.6$$

OE eg 288 / 5

A1

7

[17]

Q5.

(a) (i) $\left(x - \frac{3}{2}\right)^2$

or $p = 1.5$ stated

M1

$$\left(x - \frac{3}{2}\right)^2 + \frac{11}{4}$$

$$(x - 1.5)^2 + 2.75$$

A1

Mark their final line as their answer

2

(ii) $x = \frac{3}{2}$

correct or FT their " $x = p$ "

B1 ✓

1

(b) (i) $x^2 - 3x + 5 = x + 5 \Rightarrow x^2 = 4x$

*eliminating x or y and collecting like terms
(condone one slip)*

or $(y - 5)^2 - 3(y - 5) + 5 = y \Rightarrow y^2 - 14y + 45 = 0$

M1

$(x \neq 0) \Rightarrow x = 4$

A1

$y = 9$

A1

3

(ii) $\frac{x^3}{3} - \frac{3x^2}{2} + 5x (+c)$

one of these terms correct

M1

another term correct

A1

all correct (need not have $+c$)

A1

3

(iii) $\left[\right]_0^4 = \frac{4^3}{3} - 3 \times \frac{4^2}{2} + 5 \times 4$

must have earned M1 in part(b)(ii)

$F(\text{their } x_B) - F(0)$ "correctly sub'd"

M1

$= 17\frac{1}{3}$

$\left(\frac{64}{3} - 24 + 20 = \right) \frac{52}{3} \text{ or } \frac{104}{6} \text{ etc}$

condone 17.3 but not $16\frac{4}{3}$ etc

A1

Area trapezium $\frac{1}{2}(x_B)(5 + y_B)$

FT their numerical values of x_B, y_B

$$\text{Area} = \frac{1}{2} \times 4 \times 14 (= 28)$$

B1 ✓

$$\text{Area of shaded region } 28 - 17\frac{1}{3}$$

$$= 10\frac{2}{3}$$

$$\text{CSO; } \frac{32}{3}, \text{ accept } 10.7 \text{ or better}$$

A1

4

[13]

Q6.

$$(a) \dots\dots = \left(x^{\frac{3}{2}}\right)^2 - 2x^{\frac{3}{2}} + 1 = x^3 - 2x^{\frac{3}{2}} + 1$$

B2 for $x^3 - 2x^{\frac{3}{2}} + 1$ or $x^3 - 2x\sqrt{x} + 1$
 (B1 fully correct unsimplified expression.

seen eg $\left(x^{\frac{3}{2}}\right)^2 - x^{\frac{3}{2}} - x^{\frac{3}{2}} + 1$

or B1 for either $x^3 + 2x^{\frac{3}{2}}$... OE seen

or $x^3 + 2x^{\frac{3}{2}} - 1$ OE seen)

or B1 for $-x^3 + 2x^{\frac{3}{2}} - 1$ OE seen)

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B2,1,0

2

$$(b) \int \left(x^{\frac{3}{2}} - 1\right)^2 dx = \frac{x^4}{4} - \frac{2x^{\frac{5}{2}}}{2.5} + x (+c)$$

Ft on correct integration of all non $x^{\frac{3}{2}}$ terms
 (at least two) in c's expression. in (a)

B1F

Integration of a $kx^{\frac{3}{2}}$ as $\lambda x^{\frac{5}{2}}$ (ie power correct)

M1

$$\{= 0.25x^4 - 0.8x^{2.5} + x (+c)\}$$

Correct integration of c's $x^{\frac{3}{2}}$ term(s)
ACF

A1F

3

(c) $\int_1^4 \left(x^{\frac{3}{2}} - 1 \right)^2 dx = \left(\frac{4^4}{4} - \frac{2(4^{\frac{3}{2}})}{2.5} + 4 \right) - \left(\frac{1}{4} - \frac{2}{2.5} + 1 \right)$

$F(4) - F(1)$ attempted following integration. If $F(x)$ incorrect, ft c's answer to (b) provided integration attempted

M1

$$\{= \frac{212}{5} - \frac{9}{20} = 42.4 - 0.45\} = 41.95$$

41.95 OE eg 839 / 20
Since 'Hence' NMS scores 0 / 2

A1

2

[7]

Q7.

(a) $y = x + 3 + \frac{8}{x^4} = x + 3 + 8x^{-4}$

For $\frac{8}{x^4} = 8x^{-4}$ PI by correct differentiation
of 3rd term

B1

$$\frac{dy}{dx} = 1 - 32x^{-5} \text{ or } 1 - \frac{32}{x^5}$$

kx^{-5} OE

M1

For either

A1

3

(b) When $x = 1, y = 12$

B1

When $x = 1$, $\frac{dy}{dx} = 1 - 32 = -31$

Attempt to find value of $\frac{dy}{dx}$ when $x = 1$

M1

Tangent: $y - 12 = -31(x - 1)$

Only ft on c's answer to (a). Any correct (ft on c's (a)) form.

A1F

3

(c) $1 - 32x^{-5} = 0$

$1 - 32x^{-5} = 0$ or c's $\frac{dy}{dx} = 0$

M1

$\Rightarrow x^5 = 32$

Attempt to form $x^n = \text{const } (\neq 0)$. PI by next line

m1

$\Rightarrow x = 2$

CSO

A1

(Coordinates of M) (2, 5.5)

CSO

A1

4

(d) (i) $\int \left(x + 3 + \frac{8}{x^4} \right) dx$

$= \frac{x^2}{2} + 3x - \frac{8}{3}x^{-3} + c$

Power -3 correctly obtained

M1

$-\frac{8}{3}x^{-3}$

A1

$$\frac{x^2}{2} + 3x + c$$

B1

3

(ii)

$$= \left(2 + 6 - \frac{1}{3}\right) - \left(\frac{1}{2} + 3 - \frac{8}{3}\right)$$

Attempting to calculate $F(2) - F(1)$ where $F(x)$ is c's answer to part (d)(i) provided F is not just the c's integrand $(x + 3 + 8/x^4)$

M1

$$\frac{9}{2} + \frac{7}{3} = \frac{41}{6}$$

OE Accept 6.83 or better provided d(i) used

A1

2

(e) $k = -5.5$

Ft on $-y_M$ from part (c).

B1F

1

[16]

Q8.

(a) $(2 + x^2)^3$

$$= [(2)^3] + 3(2)^2(x^2) + 3(2)(x^2)^2 + (x^2)^3$$

For either (1), 3, 3, (1) OE unsimplified

$$\text{or } \binom{3}{1} 2^2 x^2 + \binom{3}{2} 2(x^2)^2 \text{ OE. PI}$$

M1

$$p = 3(2)^2 = 12$$

AG Be convinced. Condone left as $12x^2$

A1

$$q = 6$$

Accept left as $6x^4$

B1

3

(b) (i)
$$\int \frac{(2+x^2)^3}{x^4} dx = \int x^{-4} (8 + 12x^2 + qx^4 x^6) dx$$

or
$$\int \left(\frac{8}{x^4} + \frac{12}{x^2} + q + x^2 \right) dx$$

Uses (a) and either an indication that $\frac{1}{x^n} = x^{-n}$
in a product PI or cancelling to get at
least 3 correct ft terms

M1

$$\int (8x^{-4} + 12x^{-2} + q + x^2) dx$$

Ft on c's non-zero q. PI by next line in solution

A1F

$$= \frac{8x^{-3}}{-3} + \frac{12x^{-1}}{-1} + qx + \frac{x^3}{3} \{+c\}$$

Correct integration of either $8x^{-4}$ or $12x^{-2}$;
accept unsimplified

M1

Correct integration of both $8x^{-4}$ and $12x^{-2}$;
accept unsimplified coefficients

A1

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$$= \dots\dots\dots + 6x + \frac{x^3}{3} + c$$

$$\left(= -\frac{8}{3}x^{-3} - 12x^{-1} + 6x + \frac{x^3}{3} + c \right)$$

For $\frac{x^3}{3} + c$ simplified. The only ft is "6"
replaced by c's value for q where q is a
non-zero integer.

B1F

5

(ii)
$$\int_1^2 \frac{(2+x^2)^3}{x^4} dx = \left\{ -\frac{8}{3}(2)^{-3} - 12(2^{-1}) + 6(2) + \frac{2^3}{3} \right\} -$$

$$\left\{ -\frac{8}{3}(1)^{-3} - 12(1)^{-1} + 6(1) + \frac{1^3}{3} \right\}$$

Dealing correctly with limits; $F(2)-F(1)$ (must have attempted integration to get F ie c's F is not just the integrand)

M1

$$= \left(-\frac{1}{3} - 6 + 12 + \frac{8}{3} \right) - \left(-\frac{8}{3} - 12 + 6 + \frac{1}{3} \right)$$

$$= 16\frac{2}{3}$$

OE **exact** answer eg 50/3.

NMS scores 0

A1

2

[10]

Q9.

(a) $p = -3; q = 3$

Accept even if just embedded in the expansion

B1;B1

2

(b) (i) $\int \left(1 - \frac{1}{x^2} \right)^3 dx = \int (1 - 3x^{-2} + 3x^{-4} - x^{-6}) dx$

Uses (a) with indication of integration and

indication of $\frac{1}{x^n} = x^{-n}$ PI

M1

$$= x + 3x^{-1} - x^{-3} + \frac{1}{5}x^{-5} \{+c\}$$

At least three powers of x correctly obtained

m1

Ft on c's non-zero integers p and q .
 A1F if 3 of the 4 terms are correct (ft) or
 if all correct (ft) but left unsimplified
 Condone missing $+c$.

A2F,1F

4

(ii) $\int_{\frac{1}{2}}^1 \left(1 - \frac{1}{x^2}\right)^3 dx = \left(1 + 3 - 1 + \frac{1}{5}\right) - \left(\frac{1}{2} + 6 - 8 + \frac{32}{5}\right)$

*Attempting to calculate $F(1) - F(1/2)$
 where F is c's answer to part (b)(i)
 provided F is not the integrand or the c's*

equivalent of the integrand $\left(1 - \frac{1}{x^2}\right)^3$

M1

$$= -\frac{17}{10}$$

OE exact answer eg -1.7

A1

2

[8]

Q10.

(a) (i) $p(2) = 8 + 2 - 10$
Must find $p(2)$ NOT long division

M1

$\Rightarrow p(2) = 0 \Rightarrow (x - 2) \text{ is factor}$
Shown = 0 plus a statement

A1

2

(ii) Attempt at long division (generous)
*Obtaining a quotient $x^2 + cx + d$
 or equating coefficients (full method)*

M1

$p(x) = (x - 2)(x^2 + 2x + 5)$
 $a = 2, b = 5$ by inspection B1, B1

A1

2

(b) (i) $\frac{dy}{dx} = 3x^2 + 1$

One term correct

M1

All correct – no + c etc

A1

When $x = 2 \frac{dy}{dx} = 3 \times 4 + 1$

Sub $x = 2$ into their $\frac{dy}{dx}$

m1

Therefore gradient at Q is 13
CSO

A1

4

(ii) $y = 13(x - 2)$

*Tangent (NOT normal) attempted
ft their gradient answer from (b)(i)*

M1

CSO; correct in any form

A1

2

(iii) $\int \dots dx = \frac{x^4}{4} + \frac{x^2}{2} - 10x (+c)$

one term correct

M1

second term correct

A1

all correct (condone no +c)

A1

3

(iv) $[4 + 2 - 20] - [0] = -14$

*F(2) attempted and possibly F(0)
Must have earned M1 in (b)(iii)*

M1

Area of shaded region = 14

*CSO; separate statement following correct
evaluation of limits*

A1

2

[15]

Q11.

(a) $(n =) -4$

Accept x^{-4}

B1

1

(b) $\left(1 + \frac{3}{x^2}\right)^2 = 1 + \frac{6}{x^2} + \frac{9}{x^4}$

*Apply ISW after B2 stage
(B1 if correct but unsimplified seen)*

B2,1,0

2

(c) $\int \left(1 + \frac{3}{x^2}\right)^2 dx = x - 6x^{-1} - 3x^{-3} + c$

*At least one power of x correctly obtained
in the integration of an expansion*

M1

*A2 terms correct **and** “+ c”
(A1F two terms in x correct ft on
expansion provided integrating x to a
negative power)*

A2, 1, 0

3

(d) $\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx = \left[x - \frac{6}{x} - \frac{3}{x^3} \right]_1^3$
 $= \left(3 - \frac{6}{3} - \frac{3}{27} \right) - \left(1 - 6 - 3 \right)$

*Dealing correctly with limits; $F(3) - F(1)$
(must have attempted integration to get F)*

M1

$= 8\frac{8}{9}$

CSO;

*OE provided value is **exact**, eg $\frac{80}{9}, \frac{240}{27}$;
ISW dec value after exact value seen
NMS scores 0/2*

A1

Q12.

(a) (i) $\frac{dy}{dx} = 4 \times \frac{1}{2} x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}} = 2x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}}$

A power decreased by 1

M1

A1 For each correct term

A1 A1

3

(ii) At $P(4, 0)$, $\frac{dy}{dx} = \frac{2}{\sqrt{4}} - \frac{3}{2} \times 2$

Attempts $\frac{dy}{dx}$ when $x = 4$

M1

$= 1 - 3 = -2$

AG

A1

2

(iii) Gradient of normal = $\frac{1}{2}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 0 = m [x - 4]$

m numerical; can be awarded even if $m = -2$

M1

$y - 0 = \frac{1}{2} (x - 4) \Rightarrow 2y = x - 4$

ACF of the equation

A1

3

(iv) At Q , $x = 0$, $2y = 0 - 4$
PI

M1

$$y_e = -2$$

$$\text{Area of triangle } OPQ = 0.5 \times 4 \times |y_e|$$

Ft on a linear equation for normal provided y_e is negative and prev A1 is lost

A1F

$$= 4$$

Ft on c's negative y_e

B1F

3

$$(v) \quad 2x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 2x^{-\frac{1}{2}} = \frac{3}{2}x^{\frac{1}{2}}$$

Puts c's $\frac{dy}{dx} = 0$ and a 1st step in attempt to solve.

M1

$$2 = \frac{3}{2}x \Rightarrow x = \frac{4}{3}$$

Valid method to $ax = b$

m1

Condone 1.3 or better

A1

3

$$(b) \quad (i) \quad \int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx = 4 \frac{x^{\frac{3}{2}}}{1.5} - \frac{x^{\frac{5}{2}}}{2.5} \{+c\}$$

$$= \frac{8}{3}x^{\frac{3}{2}} - \frac{2}{5}x^{\frac{5}{2}} \{+c\}$$

One power correct

M1

Condone absence of "+ c"

A1,A1

3

$$(ii) \quad \text{Area under curve} = 4 \frac{4^{\frac{3}{2}}}{1.5} - \frac{4^{\frac{5}{2}}}{2.5} - \{0\}$$

$$F(4) - \{F(0)\}$$

M1

$$\text{Total area} = F(4) + \text{area triangle } OPQ$$

m1

$$\text{Total area} = \frac{128}{15} + 4 = \frac{188}{15} = 12.5(3\dots)$$

Accept 3 sf if clear

A1

3

[20]

Q13.

$$(a) \quad \left(1 + \frac{4}{x^2}\right)^3 = \left[1^3\right] + 3(1^2)\left(\frac{4}{x^2}\right) + 3(1)\left(\frac{4}{x^2}\right)^2 + \left[\left(\frac{4}{x^2}\right)^3\right]$$

Any valid method as far as term(s) in $1/x^2$ and term(s) in $1/x^4$

M1

$$= [1] + \frac{12}{x^2} + \frac{48}{x^4} + \left[\frac{64}{x^6}\right]$$

$$p = 12 \text{ Accept } \frac{12}{x^2} \text{ even within a series}$$

A1

$$q = 48 \text{ Accept } \frac{48}{x^4} \text{ even within a series}$$

A1

3

$$(b) \quad (i) \quad \int \left(1 + \frac{4}{x^2}\right)^3 dx = \int \left(1 + \frac{p}{x^2} + \frac{q}{x^4} + \frac{64}{x^6}\right) dx$$

*Integral of an 'expansion', at least 3 terms
PI by the next line*

M1

$$= x - px^{-1} - \frac{q}{3}x^{-3} - \frac{64}{5}x^{-5} (+c)$$

$$= x - 12x^{-1} - 16x^{-3} - \frac{64}{5}x^{-5} (+c)$$

At least two powers correctly obtained

m1

*Ft on c's non-zero integer values for p and q (A1F for two terms correct; can be unsimplified)
Condone missing c but check that signs have been simplified at some stage before the award of both A marks.*

A2F,1

4

$$(ii) \left(2 - \frac{p}{2} - \frac{q}{3(8)} - \frac{64}{5(32)} \right) - \left(1 - p - \frac{q}{3} - \frac{64}{5} \right)$$

F(2) – F(1), where F(x) is cand's answer or the correct answer to (b)(i).

M1

$$= 33.4$$

CSO

A1

2

[9]

Q14.

(a) (i) B(0, 5)

B1

Area AOB $\frac{1}{2} \times 1 \times 5$

Condone slip in number or a minus sign

M1

$$= 2 \frac{1}{2}$$

A1

3

$$(ii) \frac{3x^6}{6} + \frac{2x^2}{2} + 5x \quad \text{or} \quad \frac{x^6}{2} + x^2 + 5x \quad (\text{may have } + c \text{ or not})$$

Raise one power by 1

M1

One term correct

A1

All correct unsimplified

A1

3

- (iii) Area under curve = $\int_{-1}^0 f(x) dx$
Correctly written or $F(0) - F(-1)$ correct

B1

$$[0] - \left[\frac{1}{2} + 1 - 5 \right]$$

*Attempt to sub limit(s) of -1 (and 0)
 Must have integrated*

M1

Area under curve $3\frac{1}{2}$
CSO (no fudging)

A1

Area of shaded region = $3\frac{1}{2} - 2\frac{1}{2} = 1$
FT their integral and triangle (very generous)

B1 ✓

4

(b) (i) $\frac{dy}{dx} = 15x^4 + 2$

One term correct

M1

All correct (no +c etc)

A1

when $x = -1$, gradient = 17

CSO

A1

3

- (ii) $y = \text{"their gradient"} (x + 1)$
*Must be finding **tangent** – not normal
 any form e.g. $y = 17x + 17$*

B1 ✓

1

[14]

Q15.

(a) (i) $y = x + 1 + 4x^{-2} \Rightarrow \frac{dy}{dx} = 1 - 8x^{-3}$

M1

*Power $p \rightarrow p - 1$
(A1 if $1 + ax^n$ with $a = -8$ or $n = -3$)*

A2,1,0

3

(ii) $1 - 8x^{-3} = 0$

Puts c's $\frac{dy}{dx} = 0$

M1

$x^3 = 8$

Using $x^{-k} = \frac{1}{x^k}$ to reach $x^a = b$, $a > 0$ or correct use of logs.

m1

$x = 2$

A1

When $x = 2$, $y = 4$

A1ft

4

(iii) At (1, 6), $\frac{dy}{dx} = 1 - 8 = -7$
Attempt to find $y'(1)$

M1

Gradient of normal = $\frac{1}{7}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 6 = m[x - 1]$

m numerical

M1

$$y - 6 = \frac{1}{7}(x - 1)$$

$$\left\{ \frac{y - 6}{x - 1} = \frac{1}{7}; 7y = x + 41 \right\}$$

OE ft on c's answer for (a)(i) provided at least A1 given in (a)(i) and previous 3M marks awarded

Alft

4

(b) (i) $\int x \left(1 + \frac{4}{x^2} \right) dx =$

$$\dots\dots\dots = \frac{x^2}{2} + x - 4x^{-1} \{+c\}$$

One of three terms correct.

M1

One of three terms correct.

For A2 need all three terms as printed or better
(A1 if 2 of 3 terms correct)

A2,1,0

3

(ii) {Area = } $\int_1^4 x + 1 + \frac{4}{x^2} dx =$

$$\left[\frac{x^2}{2} + x - \frac{4}{x} \right]_1^4 = (8 + 4 - 1) - \left(\frac{1}{2} + 1 - 4 \right)$$

Dealing correctly with limits; $F(4) - F(1)$
(must have integrated)

M1

$$= 13.5$$

A1

2

[16]

Q16.

(a) (i) $\int (4 - e^{2x}) dx$

$$= 4x - \frac{1}{2} e^{2x} (+c)$$

$$4x$$

B1

$$- \frac{1}{2} e^{2x}$$

B1

2

(ii) $\int_0^{\ln 2} \left[4x - \frac{1}{2} e^{2x} \right] dx$

$$= \left[4 \ln 2 - \frac{1}{2} e^{2 \ln 2} \right] - \left[(0) - \frac{1}{2} (e^0) \right]$$

Substitute both $\ln 2$ and 0 correctly into an integrated expression

M1

$$= 4 \ln 2 - 2 + \frac{1}{2}$$

Convincing

$$= 4 \ln 2 - \frac{3}{2}$$

AG

A1

2

(b) (i) $x = 0$

$$y = 4 - 1 = \underline{\underline{3}}$$

B1

1

(ii) At B, $y = 0$

$$4 - e^{2x} = 0$$

Or reverse argument

M1

$$e^{2x} = 4$$

$$x = \ln 2$$

AG

A1

2

(c) $\frac{dy}{dx} = -2e^{2x}$

B1

$$x = \ln 2, \text{ Gradient} = -2e^{2\ln 2} = -8$$

$$x = \ln 2 \text{ into } ke^{2x}$$

M1

$$\text{Gradient normal} = \frac{1}{8} = \frac{1}{2e^{2\ln 2}}$$

OE

A1

Equation $y = -\frac{1}{8}x - \frac{1}{8}\ln 2$

OE

A1

4

(d) When $x = 0$

$$y = -\frac{1}{8}\ln 2$$

Attempt to integrate their line and substitute
 $x = 0, \ln 2$

M1

$$\text{Area } \Delta = \frac{1}{16}(\ln 2)^2 \text{ condone - ve sign} = 0.03$$

$$\frac{1}{2} (\text{their } y) \times \ln 2$$

A1 ✓

$$\text{Total area } 4\ln 2 - \frac{3}{2} + \frac{1}{16}(\ln 2)^2 = 1.30$$

AWRT

CSO

A1

3

[14]

Q17.

- (a) (i) $f(1) = 1 + 4 - 5$
must find $f(1)$ NOT long division

M1

$\Rightarrow f(1) = 0 \Rightarrow (x - 1)$ is factor
shown = 0 plus a statement

A1

2

- (ii) Attempt at $x^2 + x + 5$
*long division leading to $x^2 \pm x \dots$ or
 equating coefficients*

M1

$f(x) = (x - 1)(x^2 + x + 5)$
 $p = 1, q = 5$ by inspection scores B1, B1

A1

2

- (iii) $(x =)1$ is real root

B1

Consider $b^2 - 4ac$ for their $x^2 + x + 5$
not the cubic!

M1

$$b^2 - 4ac = 1^2 - 4 \times 5 = -19 < 0$$

Hence no real roots (or only real root is 1)
CSO; all values correct plus a statement

A1

3

- (b) (i) $\int \dots dx = \frac{x^2}{4} + 2x^2 - 5x (+c)$
one term correct unsimplified

M1

second term correct unsimplified

A1

all correct unsimplified

A1

3

(ii) $[4 + 8 - 10] - \left[\frac{1}{4} + 2 - 5 \right]$

*correct use of limits 1 and 2;
F(2) – F(1) attempted*

M1

$$= 4 \frac{3}{4}$$

A1

Area of $\Delta = \frac{1}{2} \times 11 = 5 \frac{1}{2}$
correct unsimplified

B1

$\Rightarrow$ shaded area $= 5 \frac{1}{2} - 4 \frac{3}{4}$

*combined integral of $7x - 6 - x^3$ scores
M1 for limits correctly used then*

$$= \frac{3}{4}$$

A3 correct answer with all working correct

A1

4

[14]

Q18.

(a) (i) x^2

B1

1

(ii) $x^{\frac{1}{2}} = \sqrt{x}$

Accept either form

B1
1

(iii) x^3

B1
1

(b) (i) $\int 3x^{\frac{1}{2}} dx = \frac{3}{\frac{3}{2}} x^{\frac{3}{2}} \{+c\}$

Index raised by 1

M1

Simplification not yet required

A1

$= 2x^{\frac{3}{2}} + c$

Need simplification and the + c OE

A1
3

(ii) $\int_1^9 3x^{\frac{1}{2}} dx = (2 \times 9^{\frac{3}{2}}) - (2 \times 1^{\frac{3}{2}})$

F(9) – F(1), where F(x) is candidate's answer to (b)(i) [or clear recovery]

M1

$= 52$

Ft on (b)(i) answer of form $kx^{1.5}$ i.e. $26k$

A1ft
2

[8]

Q19.

(a) $y_D = 3 + 1 = 4$ or $y_C = 12 - 8 = 4$

Attempt at either y coordinate

M1

$\text{Area } ABCD = 3 \times 4 = 12$

A1
2

(b) (i) $x^3 - \frac{x^4}{4} \quad (+ C)$

Increase one power by 1

M1

One term correct unsimplified

A1

All correct unsimplified (condone no +C)

A1

3

(ii) Sub limits -1 and 2 into their (b)(i) ans

May use both -1, 0 and 0, 2 instead

M1

$$[8 - 4] - \left[-1 - \frac{1}{4} \right] = 5 \frac{1}{4}$$

A1

Shaded area = "their" (rectangle - integral)

Alt method: difference of two integrals

M1

$$= 12 - 5 \frac{1}{4} = 6 \frac{3}{4}$$

CSO. Attempted M2, A2

A1

4

(c) (i) $\frac{dy}{dx} = 6x - 3x^2$

One term correct

M1

All correct (no +C etc)

A1

2

(ii) When $x = 1$, $y = 2$ when $x = 1$,

May be implied by correct tgt equation

B1

$$\frac{dy}{dx} = 3 \text{ as 'their' grad of } \tan$$

For their derivative when $x = 1$

M1ft

Tangent is $y - 2 = 3(x - 1)$
Any correct form $y = 3x - 1$ etc

A1

3

(iii) Decreasing when $\frac{dy}{dx} = 6x - 3x^2 < 0$
Watch no fudging here!!
May work backwards in proof.

M1

$$3(2x - x^2) < 0 \Rightarrow x^2 - 2x > 0$$

AG (be convinced no step incorrect)

A1

2

- (d) Two critical points 0 and 2
Marked on diagram or in solution

M1

$x > 2$, $x < 0$ ONLY
or M1 A0 for $0 < x < 2$ or $0 > x > 2$
SC B1 for $x > 2$ (or $x < 0$)

A1

2

[18]

Q20.

(a) $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - 3$
One term correct

M1

Both correct

A1

2

- (b) (i) When $x = 0$, $\frac{dy}{dx} = -3$
Ft provided answer < 0.

B1F ✓

Eqn of tangent at O is $y = -3x$
OE Ft on $y'(0)$

B1F ✓

2

- (ii) At $(9,0)$ $\frac{dy}{dx} = \frac{3}{2}(9)^{\frac{1}{2}} - 3$
Attempt to find $y'(9)$

M1

Eqn tangent at A is $y - 0 = y'(9)[x - 9]$
OE

m1

$$\Rightarrow y = \frac{3}{2}(x - 9) \Rightarrow 2y = 3x - 27$$

CSO. AG

A1

3

- (iii) Eliminating $y \Rightarrow -6x = 3x - 27$
OE method to one variable
(eg $2y = -y - 27$)

M1

$$9x = 27 \Rightarrow x = 3$$

*[A1F for each coordinate; only ft on $y = kx$ tangent
 in (b)(i) for $k < 0$]*

A1F

When $x = 3$, $y = -9$. $P(3, -9)$

A1F

3

(c) $\int \left(x^{\frac{3}{2}} - 3x \right) dx = \frac{2}{5} x^{\frac{5}{2}} - \frac{3x^2}{2} (+c)$

One power correct

M1

Condone absence of "+ c" and unsimplified forms

A2,1,0

3

(d) $\int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx =$
PI

B1

$$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{3}{2} \times 9^2 - 0$$

Correct use of limits following integration

M1

$$= -24.3$$

Area of triangle $OPA = \frac{1}{2} \times 9 \times |y_P|$

M1

Sh.Area = $\frac{1}{2} \times 9 \times |y_P| - \left| \int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx \right|$
OE

M1

$$= 40.5 - 24.3 = 16.2$$

A1

5

[18]

Q21.

(a) (i) $\frac{dy}{dx} = 3x^2 - 20x + 28$

One term correct

M1

Another term correct

A1

All correct (no + c etc)

A1

3

- (ii) Their $\frac{dy}{dx} = 0$ for stationary point
Or realising condition for stationary point

M1

$(x - 2)(3x - 14) = 0$
Attempt to solve using formula/ factorise

m1

$\Rightarrow x = 2$

A1

or $x = \frac{14}{3}$

*Award M1, A1 for verification that $x = 2 \Rightarrow \frac{dy}{dx} = 0$
 then may earn m1 later*

A1

4

- (b) (i) $\frac{x^4}{4} - \frac{10x^3}{3} + 14x^2$ (+ c)
One term correct unsimplified

M1

Another term correct unsimplified

A1

All correct unsimplified (condone missing + c)

A1

3

- (ii) $\left[\frac{81}{4} - 90 + 126 \right] \quad (-0)$
Attempt to sub limit 3 into their (b)(i)

M1

$= 56\frac{1}{4}$

AG Integration, limit subtraction all correct

A1

2

(iii) Area of triangle = $31 \frac{1}{2}$

Correct unsimplified $\frac{1}{2} \times 21 \times 3$

B1

Shaded Area = $56 \frac{1}{4}$ – triangle area

M1

= $24 \frac{3}{4}$

Or equivalent such as $\frac{99}{4}$

B1

3

[15]

Q22.

(a) $y = x \ln x$
 $\frac{dy}{dx} = x \times \frac{1}{x} + \ln x$

use of product rule (only differentiating, 2 terms with + sign)

M1

= $\ln x + 1$

A1

2

(b) $\int (\ln x + 1) dx = x \ln x$

OE; attempt at parts with

M1

$\int \ln x dx = x \ln x - x (+c)$

A1

2

(c) $\int_1^5 \ln x \, dx = [x \ln x - x]_1^5$

$= (5 \ln 5 - 5) - (1 \ln 1 - 1)$

correct substitution of limits into their (b) provided $\ln x$ is involved

M1

$5 \ln 5 - 4$

ISW

A1

2

[6]

Q23.

- (a) (i) $x - 1$ is a factor

May be earned retrospectively

B1

$x + 2$ is a factor

From part (iii)

B1

2

- (ii) Attempt at third factor

Multiplying/ dividing/ factor theorem

M1

$f(x) = (x - 1)(x + 2)(x - 4)$

$(x+4) \Rightarrow M1, A0$

A1

2

- (b) (i) At A, $y = 8$

Or (0,8)

B1

1

- (ii) At B, $x = 4$

Or (4,0) NO ft of wrong factor

B1

1

(c) (i) $\frac{x^4}{4} - x^3 - 3x^2 + 8x (+c)$

Increase one power by 1

M1

One term correct (unsimplified)

A1

Two other terms correct (unsimplified)

A1

*All correct (unsimplified)
(condone missing + c)*

A1

4

(ii) Realisation that limits are -2 and 1

Condone wrong way round

B1

$$\text{Area} = \left[\frac{1}{4} - 1 - 3 + 8 \right] - [4 + 8 - 12 - 6]$$

Attempt to sub their limits into their (c)(i)

M1

$$= 20 \frac{1}{4}$$

CSO. Must use $F(1) - F(-2)$ correctly

A1

3

[13]

Q24.

(a) $\sqrt{x} = x^{\frac{1}{2}}$

Accept $k = 0.5$

B1

1

(b) $\sqrt{x}(x-1) = x^{\frac{1}{2}}x - x^{\frac{1}{2}} = x^{\frac{3}{2}} - x^{\frac{1}{2}}$

M1

Accept $p = 1.5, q = 0.5$

A1

2

(c) $\int \sqrt{x}(x-1) dx = \frac{x^{2.5}}{2.5} - \frac{x^{1.5}}{1.5} (+c)$
Increases a power of x by 1

M1

ft non-integer p

A1ft

ft non-integer q

A1ft

3

(d) $\int_1^2 dx = \left(\frac{2^{2.5}}{2.5} - \frac{2^{1.5}}{1.5} \right) - \left(\frac{1}{2.5} - \frac{1}{1.5} \right)$
Limits; $F(2) - F(1)$

M1

$\dots = \left(\frac{4\sqrt{2}}{2.5} - \frac{2\sqrt{2}}{1.5} \right) - \left(\frac{1}{2.5} - \frac{1}{1.5} \right)$
Fractional powers to surds

m1

$\left(\frac{24\sqrt{2}}{15} - \frac{20\sqrt{2}}{15} \right) - \left(-\frac{4}{15} \right) = \text{pr. ans}$
CSO AG (be convinced)

A1

3

[9]

Q25.

(a) (i) $p(3) = 27 - 45 + 21 - 3$
Finding $p(3)$

M1

$p(3) = 0 \Rightarrow x - 3$ is a factor
Shown = 0 plus a statement
 Or $(x - 3)(x^2 - 2x + 1)$

A1
2

- (ii) B is point (3,0)
Must have coordinates

B1
1

(b) (i) $\frac{dy}{dx} = 3x^2 - 10x + 7$
One term correct

M1

All correct with NO + c etc

A1
2

(ii) $3x^2 - 10x + 7 = 0$

Putting their $\frac{dy}{dx} = 0$

M1

$\Rightarrow (x - 1)(3x - 7) = 0$
Attempt to use quad formula or factorise

m1

$\Rightarrow$ at M , $x = \frac{7}{3}$
CSO factors correct etc

A1
3

(c) $\frac{d^2y}{dx^2} = 6x - 10$

ft their $\frac{dy}{dx}$

B1ft

sub $x = 1$, $\Rightarrow \frac{d^2y}{dx^2} = -4$
ft their $\frac{d^2y}{dx^2}$

B1ft

2

(d) (i) $\frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x$ (+c)
Increase one power by 1

M1

One term correct

A1

Two other terms correct

A1

All correct (condone missing + c)

A1

4

- (ii) Realisation that limits are 0 and 1
Condone wrong way round

B1

$$\left[\frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3 \right] - 0$$

Attempt to sub their limits into their (d)(i)

M1

$$= -\frac{11}{12}$$

CSO. Must use $F(1) - F(0)$ correctly

A1

$$\text{Area} = \frac{11}{12}$$

CSO. Convincing argument

E1

4

[18]

Q26.

(a) (i) $\sqrt{x} = x^{\frac{1}{2}}$
Accept $p = 0.5$

B1

1

(ii) $\int \sqrt{x} \, dx = \frac{x^{1.5}}{1.5} \{+c\}$
Index raised by 1

M1

Correct ft on p. Condone missing '+c'

A1ft

2

(iii) $\text{Area} = \int_1^4 \sqrt{x} \, dx$
Limits 1 and 4 PI

B1

..... = $\frac{4^{1.5}}{1.5} - \frac{1}{1.5}$
F(4) – F(1)

M1

= $\frac{14}{3}$

Accept 4.66 or better

A1

3

(b) (i) $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$
Index (p – 1) ft

M1

When $x = 4$, $y'(4) = 0.25$
Attempt to find $y'(4)$

M1

When $x = 4$, $y = 2$

B1

Equation of tangent: $y - 2 = \frac{1}{4} (x - 4)$

accept other forms

A1

4

- (ii) When $x = 0, y = 1$ $B(0, 1)$
Subs $x = 0$ and then $y = 0$ into equation of tangent. PI

M1

When $y = 0, x = -4$ $A(-4, 0)$
*Correct ft y_B and x_A
 (may be awarded as part of area calculation)*

A1ft

Area = $0.5(1)(4) = 2$
ft wrong sloping tangent and max of 1 further slip. Final answer must be +ve

A1ft

3

- (c) Translation
'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided linked to 'translation/move/shift' (Note: B0B1 is possible)

B1

2

- (d) $h = 1$

PI

B1

Integral = $h/2 \{ \dots \}$
 $\{ \dots \} = f(1) + 2[f(2) + f(3)] + f(4)$
OE summing of areas of the three traps

M1

$\{ \dots \} = 0 + 2(1 + \sqrt{2}) + \sqrt{3}$
Condone 1 numerical slip

A1

$$\text{Integral} = \frac{1}{2} \{2(1 + 1.414\dots) + 1.732\}$$

$$\text{Integral} = 0.5 \times 6.560\dots = 3.28 \text{ to 3sf}$$

Must be 3.28

A1CAO

4

[19]

Q27.

(a) $y = x \sin 2x$

$$\frac{dy}{dx} = x 2\cos 2x + \sin 2x$$

M1

product rule

A1,A1

3

(b) (i) $y = (x^2 - 6)^4$

$$\frac{dy}{dx} = 4(x^2 - 6)^3 (2x) \quad (\text{or better})$$

M1 for $(x^2 - 6)^3$

M1A1

2

(ii) $\int 8x (x^2 - 6)^3 dx = (x^2 - 6)^4$

for c $(x^2 - 6)^4$ if correct attempt

M1

$$\int = \frac{1}{8} (x^2 - 6)^4 (+c)$$

A1

for $\frac{1}{k} (x^2 - 6)^4$ at 'by parts' M1A0

for k = 8

OR

$$(x^2 - 6)^3 = x^6 - 18x^4 + 108x^2 - 216 \quad (\text{M1A1})$$

$$\int x (x^2 - 6)^3 = \frac{x^8}{8} - 3x^6 + 27x^4 - 108x^2 \quad (\text{A1})$$



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